PORTFOLIO

I’m Sairaj Walake I don’t just solve problems, I redesign them. My journey began with preparing for JEE, where I developed analytical thinking, but I soon realized my true strength lay in creativity and hands on exploration. Since childhood, I’ve been driven by curiosity, often taking things apart to understand how they work and rebuilding them into something new and improved. This natural instinct to question, experiment, and innovate gradually led me toward the field of design.

Beyond academics, I am a sports enthusiast with achievements in badminton and a strong foundation in martial arts and self-defense. These experiences have shaped my discipline, resilience, and focus, influencing the way I approach challenges. I believe in observing deeply, thinking critically, and executing ideas with precision.

I aim to create designs that are not only functional but also meaningful solutions that connect with people and improve everyday experiences. By blending creativity with structure and curiosity with discipline, I strive to continuously grow and make a lasting impact through design.

Project Background

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FlockFeeder

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1 FLOCK FEEDER

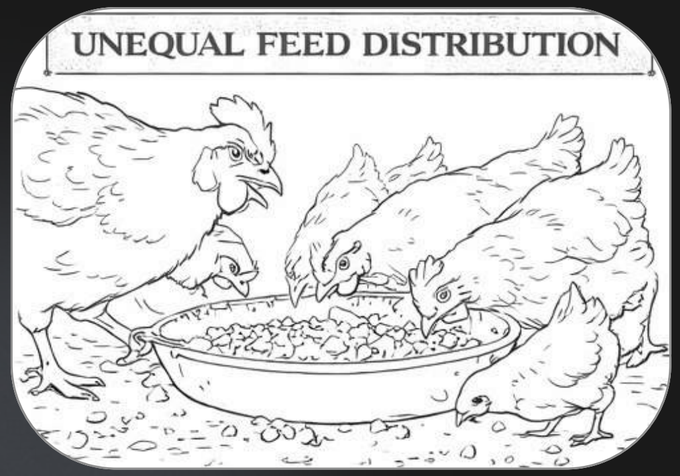
Poultry flock feeders have evolved alongside the growth of farming practices. In early times, chickens were raised in backyards and fed using simple containers like pots, tins, or wooden troughs. These methods often caused feed wastage, contamination, and access by pests.

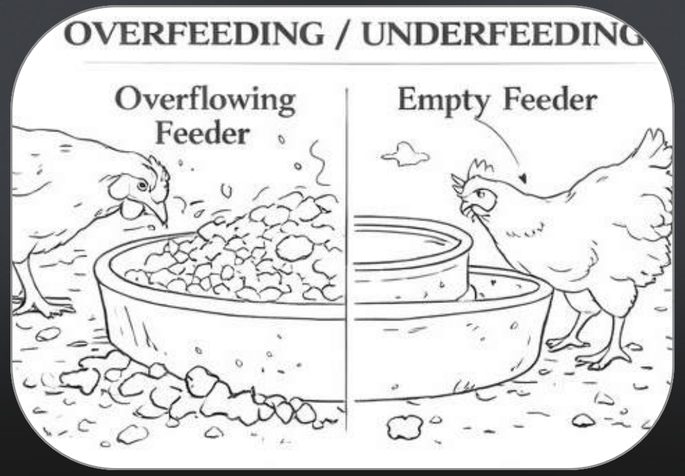
As poultry farming expanded in the late 19th and early 20th centuries, more efficient feeders such as gravity-fed and hanging designs were introduced. These helped reduce waste and improved feeding organization, though manual effort was still required.

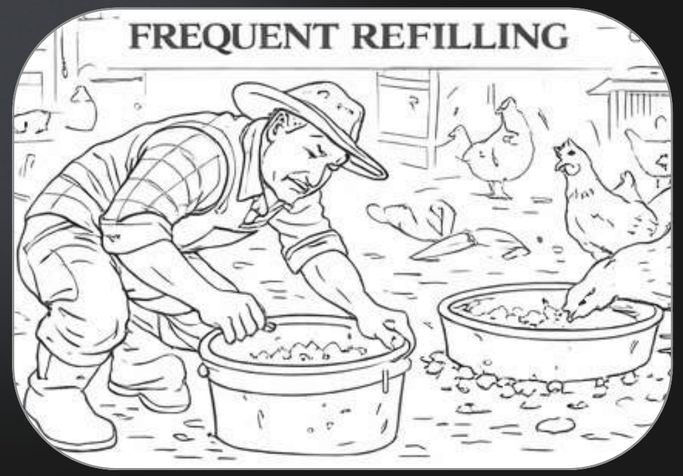
The mid-20th century marked a major advancement with the introduction of automatic feeding systems. Chain-driven and centralized feeders allowed uniform feed distribution across large flocks, reducing labor and improving productivity.

Today, modern flock feeders use automation, sensors, and precision systems to control feed quantity and maintain hygiene. They are designed to minimize waste, prevent contamination, and support animal welfare, reflecting the shift toward efficient, large-scale poultry farming.

System Challenges

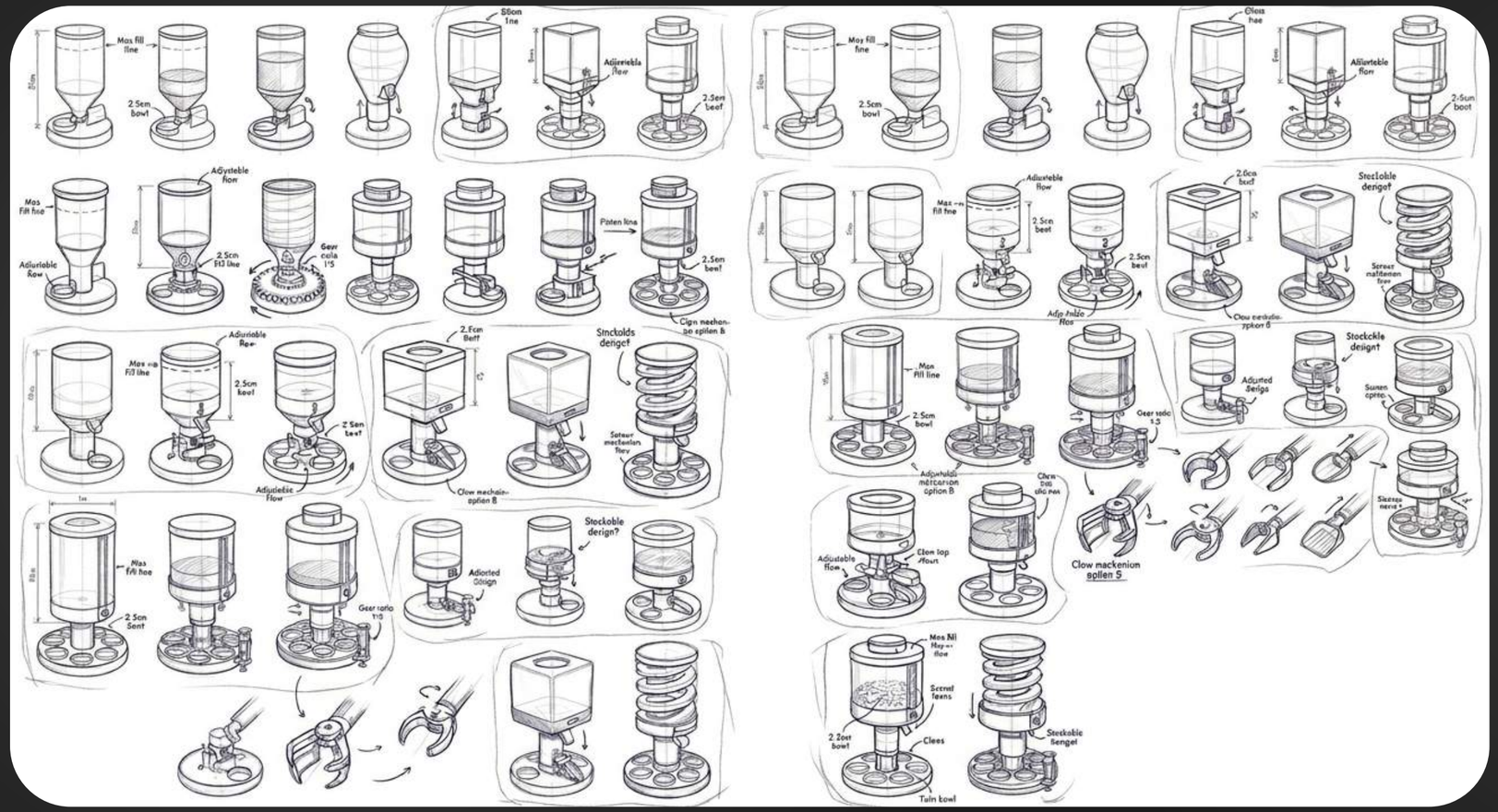






Visual Direction

Shaping Ideas



Concept model

This poultry feeder is designed as an efficient and hygienic feeding system that ensures controlled distribution of feed. The transparent hopper allows easy monitoring of feed levels, while the feed flow control mechanism regulates the quantity dispensed to prevent wastage. The central column directs feed downward into multiple stainless steel feeding trays, enabling simultaneous access for several birds.

A compression spring and adjustment knob system helps in controlling feed release based on demand, improving usability and reducing manual intervention. The use of stainless steel components ensures durability, corrosion resistance, and easy cleaning, while the plastic outer body keeps the structure lightweight and cost-effective.

Overall, the design focuses on functionality, cleanliness, and efficient feed management in poultry farming.

Exploded view

Multi View Analysis

Top view

SIDE / FRONT VIEW

Bottom view

Process Documentation